

“ WE OUGHT NOT TO SAY THAT BECAUSE THE **EARTH IS SMALLER THAN THE SUN** AND IS INFLUENCED BY THE SUN, IT IS **MORE LOWLY.** ”

Nicholas of Cusa, German philosopher, in *De Docta Ignorantia*, 1440



An early 16th-century woodcut shows Nicholas of Cusa caught between a group advocating church reform (as Nicholas did) and conservative Papal supporters.



JOHANNES GUTENBERG [c.1400–68]

Born in Mainz, Johannes Gutenberg later moved to Strasbourg where he engaged in a mysterious venture he called “adventure and art”—perhaps his first experiments in printing. By 1448, he was back in Mainz, where, by 1450, his printing press was in operation. The venture did not prosper and by 1459 Gutenberg was bankrupt.

Printing for the masses

This replica of Gutenberg’s press shows the type of machinery that he employed in Mainz for the production of his 42-line Bible, so called because of its 42-line columns.

cosmological ideas. He held the radical view that **Earth rotates on its own axis and orbits around the Sun**, prefiguring Copernicus’s theory 100 years later.

Around 1440, **Johannes Gutenberg** began experiments with **printing using movable type**. The blocks could be moved as required and later reused. By

1450, he had established a **printing press** that produced the earliest extant printed work in Europe, an edition of the *Ars Grammatica*. Gutenberg’s printing techniques grew more sophisticated and, in 1454, he published an edition of the Bible.

“ IT IS A **PRESS...** FROM WHICH SHALL FLOW IN INEXHAUSTIBLE STREAMS... **LIKE A NEW STAR IT SHALL SCATTER THE DARKNESS OF IGNORANCE.** ”

Johannes Gutenberg, German printer, c.1450



1430–40
Al-Qalasadi publishes work using symbols for algebraic operations

1436 Leon Battista Alberti's *On Painting* describes mathematical principles of perspective in painting

1437 Astronomers at Samarkand observatory publish *Zij-i-Sutani* star catalog

c.1440 Gutenberg begins experiments with printing using movable type

1440 Nicholas of Cusa proposes that Earth moves around the Sun, and that other stars may have inhabited planets

1436 Brunelleschi completes the dome of Florence Cathedral